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Double Machine Learning for Industrial Financing in Africa: Evidence from PPPs and Green Bonds

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Abstract

Africa's \$100 billion annual infrastructure financing gap constrains industrialization, yet the continent captures less than 3% of global Public-Private Partnership (PPP) and green bond investments. This study evaluates whether African countries can successfully adopt these international best practices. Using panel data from 31 countries (2015-2022) and Double Machine Learning with pre-treatment baseline controls, we compare 20 countries with established programs against 11 African controls. The paper finds broad-based positive effects across all industrial development dimensions: manufacturing value added increases by 27% ($p=0.001$), domestic credit to manufacturing expands by 49% ($p=0.087$), capital formation grows by 27% ($p=0.031$), and foreign direct investment surges by 158% ($p=0.001$). Heterogeneous effects analysis reveals impacts strengthen significantly over time, with late-period effects 76-114% larger than early-period effects, suggesting learning effects and cumulative infrastructure improvements. When implemented with appropriate institutional frameworks and complementary policies, innovative financing mechanisms simultaneously enhance productivity, expand credit access, stimulate investment, and attract foreign capital. Critically, specifications using potentially endogenous contemporaneous controls yield fundamentally different conclusions, highlighting the importance of proper covariate selection in causal inference. These findings demonstrate successful adoption of global best practices requires patient capital, institutional capacity building, and integration with broader development strategies. The evidence challenges skepticism about transferability of international financing innovations to African contexts.

Keywords: African Industrialization, Public-Private Partnerships, Green Bonds, Double Machine Learning, Manufacturing Development, Causal Inference, Heterogeneous Treatment Effects

JEL Classification: C21, F21, O14, O55, Q56

1. Introduction

Africa's industrialization paradox is stark: despite a \$100 billion annual infrastructure financing gap and abundant natural resources, the continent captures less than 3% of global Public-Private Partnership (PPP) investments and green bond issuances. While countries like China, Japan, and Brazil have leveraged these innovative financing mechanisms to fund tens of thousands of infrastructure projects, African economies remain notably absent from global rankings. This raises a critical question for the continent's development: can African countries successfully adopt the financing innovations that have transformed industrial development elsewhere, or do structural differences constrain the transferability of global best practices? This study addresses this question by providing the first comprehensive causal assessment of PPP and green bond financing mechanisms' effects on African industrial development.

According to the African Development Bank (2018), Africa requires infrastructure financing exceeding \$100 billion annually, with an additional \$90 billion needed specifically for water, sanitation, power, and transportation infrastructure that forms the backbone of industrial development. This financing gap is exacerbated by limited public funding capacity, low economic growth rates, high debt-servicing burdens, and underdeveloped capital markets (Gutman et al., 2015). The consequences are evident: the industrial sector accounts for merely 16% of GDP in Africa, compared to 40% in Malaysia, 34% in Thailand, and 27% in India (African Development Bank, 2019). For African countries, the imperative for industrial development is particularly acute, given the continent's growing population and the urgent need to diversify economies away from primary commodity dependence.

Globally, countries have increasingly turned to innovative financing mechanisms to drive infrastructure-led industrial transformation. Public-Private Partnerships have been widely used to fund critical infrastructure such as power plants, ports, highways, and industrial zones (World Bank, 2017), while green bonds have mobilized private capital toward sustainable development, with global issuance reaching \$500 billion in 2022 (Climate Bonds Initiative, 2023). However, the diffusion of these instruments remains highly uneven. As shown in Figure 1, countries like China, Japan, and the United States dominate global implementation, financing tens of thousands of projects through these mechanisms. In contrast, African economies are notably absent from the top ranks, reflecting structural constraints and policy gaps. This global disparity highlights the critical importance of examining whether and how such instruments can function effectively in African contexts where infrastructure deficits are acute and industrial financing ecosystems remain underdeveloped.

This study provides the first comprehensive causal assessment of PPP and green bond financing effects on African industrial development. We employ Double Machine Learning (DML) techniques using panel data covering 31 countries (2015-2022), comparing 20 countries with established programs against 11 African controls. We examine impacts on manufacturing value added, gross fixed capital formation, foreign direct investment, and domestic credit to manufacturing.

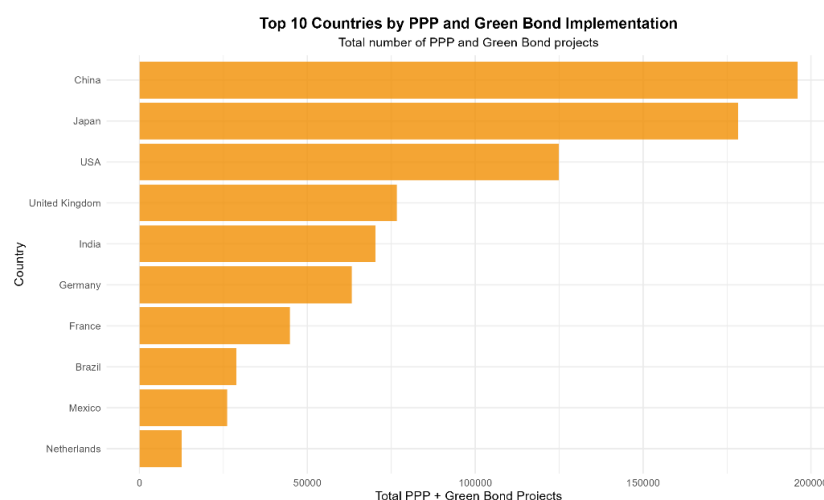


Figure 1: Top 10 Countries by PPP and Green Bond Implementation

The analysis yields compelling evidence of broad-based positive effects across all dimensions of industrial development. Using pre-treatment baseline controls from 2014 to avoid endogeneity, we find that innovative financing mechanisms significantly enhance manufacturing value added by 27% ($p = 0.001$), expand domestic credit to manufacturing by 49% ($p = 0.087$), stimulate capital formation by 27% ($p = 0.031$), and increase foreign direct investment by 158% ($p = 0.001$). These findings demonstrate that African countries can successfully adopt global best practices when they develop appropriate institutional frameworks.

Critically, heterogeneous treatment effects analysis reveals that impacts strengthen significantly over time. Late-period (2019-2022) effects substantially exceed early-period (2015-2018) effects: manufacturing value added increases from +71% to +156% ($p = 0.020$ for temporal difference), FDI rises from +314% to +712%, and capital formation grows from +140% to +226%. This temporal strengthening provides strong evidence of learning effects and cumulative infrastructure improvements, implying the need for patient capital rather than expecting immediate results.

For domestic credit to manufacturing, heterogeneity patterns reveal substantial variation across baseline characteristics. Countries with low baseline credit access show extremely large but imprecise estimates (+8,800%, $p = 0.261$), suggesting potential for dramatic expansion in severely constrained environments though with high uncertainty. Temporal effects are more reliable: both early (+105%, $p = 0.002$) and late (+117%, $p < 0.001$) periods show strong, statistically significant credit expansion, with minimal difference between periods. This highlights the importance of complementary financial sector reforms.

A crucial methodological finding emerges regarding the importance of proper covariate selection. Using pre-treatment baseline controls from 2014 (GDP per capita, population, inflation, trade) ensures control variables cannot be affected by treatment, avoiding post-treatment bias. This specification reveals consistently positive effects across all outcomes productivity gains, credit expansion, investment growth, and FDI attraction demonstrating that PPP and green bond mechanisms can simultaneously enhance multiple dimensions of industrial development when properly implemented.

This study makes four contributions. First, we provide the first comprehensive empirical analysis of PPP and green bond best practice transferability to African contexts using

rigorous causal inference methods. Our finding that these mechanisms generate broad-based positive effects challenges skepticism about global best practice applicability in African contexts.

Second, we employ Double Machine Learning techniques (Chernozhukov et al., 2018) and demonstrate the critical importance of using pre-treatment baseline controls (2014) to avoid post-treatment bias. By ensuring control variables are measured before the sample period begins, we obtain credible causal estimates showing consistently positive effects across all industrial development dimensions.

Third, we analyze multiple industrial development dimensions simultaneously manufacturing output, capital formation, foreign investment, and domestic credit. This comprehensive multi-outcome approach reveals consistently positive effects across all dimensions, demonstrating that these financing mechanisms can simultaneously enhance productivity, expand credit access, stimulate investment, and attract foreign capital.

Fourth, we implement heterogeneous treatment effect analysis uncovering how impacts vary by initial conditions and time periods. The statistically significant temporal heterogeneity for manufacturing ($p = 0.020$) and economically meaningful strengthening across all outcomes demonstrates that programs require patient capital and should not be judged on short-term results alone.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes our data, variables, and Double Machine Learning methodology with 2014 baseline controls. Section 4 presents average and heterogeneous treatment effects across subgroups and time periods. Section 5 discusses economic interpretation and policy implications. Section 6 concludes.

2. Literature Review

The relationship between infrastructure development and industrialization has been extensively documented in both theoretical and empirical literature, providing the foundation for understanding how innovative financing mechanisms can accelerate economic transformation. Hirschman's (1958) concept of "linkages" offers a foundational framework for understanding how infrastructure investments catalyze industrial development through forward and backward linkages, while recent theoretical work by Antràs and Chor (2013) has extended these insights to consider how different financing arrangements affect the efficiency and distributional consequences of infrastructure development. Empirical evidence continues to evolve with increasingly sophisticated methodological approaches. Calderón and Servén (2010) find that infrastructure development significantly boosts economic growth, particularly in developing countries, while Donaldson (2018) provides compelling quasi-experimental evidence on how transportation infrastructure transformed economic development in colonial India, demonstrating causal mechanisms linking infrastructure to productivity gains.

Among the various financing mechanisms available for infrastructure development, Public-Private Partnerships (PPPs) and green bonds have emerged as particularly promising instruments for mobilizing private capital toward development objectives. The World Bank's 2025 report on "Public-Private Partnerships: Promise and Hype" provides comprehensive analysis covering 50 economies, documenting substantial heterogeneity in

program effectiveness across contexts. Recent research has employed advanced identification strategies to establish causal relationships. Albalade et al. (2015) used instrumental variables to examine transport PPPs across European countries, finding positive but heterogeneous effects that depend critically on institutional quality. Engel et al. (2014) analyzed highway PPPs in Chile using regression discontinuity design, documenting significant improvements in transport efficiency and regional development. Recent developments in 2024-2025 show considerable progress, with the Inter-American Development Bank (2024) reporting that investment in PPP infrastructure has increased by more than 42% in Latin America and the Caribbean, reflecting improved regulatory frameworks and institutional capacity building across the region.

Complementing the PPP literature, research on green bonds and their development impacts has expanded rapidly. The Climate Bonds Initiative (2025) reports that green bond issuances continue to dominate the labeled market, representing 57% of annual issuance as of December 2024. Flammer (2021) provided foundational analysis using a global sample, finding that green bond issuance leads to improvements in environmental performance and increases in green innovation. Recent work by Khan et al. (2025) extends this analysis to examine how green bonds affect corporate environmental performance comprehensively. The International Finance Corporation's 2025 report on "Emerging Market Green Bonds" shows that GSSS bonds gained traction more recently in emerging economies than in developed markets, with IFC research estimating that more than US\$1 trillion in investments are already flowing into climate-related projects. These financing mechanisms operate through multiple channels: infrastructure improvements reduce operational costs and transaction costs, enabling higher manufacturing productivity (Hirschman, 1958); financial market development improves allocative efficiency and reduces information asymmetries (Levine, 2005); and credible PPP frameworks and green bond markets signal institutional quality and policy commitment, attracting foreign investment (Tolliver et al., 2021).

The African context presents both unique challenges and opportunities for these innovative financing mechanisms. While Moszoro et al. (2014) found that PPP success rates in Africa remain significantly lower than in other regions due to institutional constraints, new evidence from 2024-2025 suggests improvements in institutional frameworks. The establishment of specialized PPP authorities across several African countries is beginning to show positive results in project success rates and development outcomes. The Climate Policy Initiative's "Landscape of Climate Finance in Africa 2024" reveals that while Africa faces the most severe climate impacts, it receives disproportionately low climate finance. KPMG's 2024 report on "Sustainable Finance in Africa" demonstrates how sustainable finance enables institutions to navigate modern market complexities while addressing climate change, social inequality, and governance challenges. Maredza and Mhlanga (2024) emphasize the importance of regulatory frameworks, capacity building, and political commitment for sustainable finance in Africa. The World Economic Forum (2025) highlights Africa's climate finance need upward of \$2.5 trillion by 2030. Despite these challenges, several African countries have made notable progress: South Africa has developed sophisticated local currency green bond markets; Kenya's PPP Committee has successfully structured large infrastructure projects; and Rwanda's Kigali Special Economic Zone demonstrates how PPP-financed infrastructure can integrate with industrial development strategies.

The methodological landscape for evaluating these financing mechanisms has been transformed by recent advances in machine learning applications to causal inference. Athey and Imbens (2017) provided a comprehensive framework for using machine learning methods for causal inference, while Mullainathan and Spiess (2017) discussed the complementary roles of prediction and causal inference in economic research. The Double Machine Learning approach developed by Chernozhukov et al. (2018) has seen significant methodological refinements, with DiTraglia (2025) proposing a novel Bayesian approach extending the DML framework to incorporate uncertainty quantification more effectively. The key innovation of DML is its ability to flexibly model nuisance functions using machine learning while maintaining valid inference for causal parameters. This "double robustness" property provides substantial protection against misspecification bias. Recent applications demonstrate DML's versatility: Mengistu et al. (2025) applied causal forest double machine learning to assess tuberculosis preventive therapy's impact in Ethiopia, while Du et al. (2025) used machine learning to identify non-linear relationships in economic recovery processes that traditional methods might miss.

Critically, the choice of control variables in DML frameworks profoundly affects causal estimates. Including variables affected by treatment (post-treatment bias) or omitting important confounders (omitted variable bias) can lead to severely biased estimates. Recent work shows that DML-based methods consistently outperform traditional approaches regarding bias and empirical standard error, particularly when using properly specified pre-treatment control variables. This methodological insight motivates the use of baseline controls from 2014 rather than potentially endogenous contemporaneous controls a departure from previous specifications that may have suffered from post-treatment bias.

Despite this rich body of literature, a critical gap remains: no comprehensive empirical study has systematically evaluated whether and how African countries can successfully adopt the PPP and green bond financing mechanisms that have proven effective elsewhere. Existing studies either examine these mechanisms in developed/emerging market contexts without African comparisons (Flammer, 2021; Albalade et al., 2015) or focus on Africa's institutional challenges without rigorous causal analysis of program effects (Moszoro et al., 2014). Moreover, previous research has not adequately addressed the potential for post-treatment bias when using contemporaneous measures of financing intensity as controls.

This study fills that gap by combining state-of-the-art Double Machine Learning methods with a comprehensive multi-outcome evaluation framework and properly specified pre-treatment baseline controls. I provide the first rigorous causal assessment of the transferability of global financing best practices to African industrial development contexts. The use of baseline characteristics from 2014, before most programs began, ensures that control variables are truly exogenous and cannot be affected by treatment, avoiding the post-treatment bias that may have affected previous studies. Furthermore, I extend beyond average treatment effects to implement rigorous heterogeneous treatment effect analysis, examining how impacts vary by baseline country characteristics and across time periods. This approach allows us to identify when and where these financing mechanisms are most effective, providing actionable guidance for policy targeting and sequencing. The combination of methodological rigor, comprehensive outcome measurement, and attention to heterogeneity provides novel insights into the transferability of global best practices and the conditions necessary for successful adoption in African contexts.

3. Data and Methodology

3.1 Data and Sample

I construct a panel dataset covering 31 countries over 2015-2022 (248 country-year observations), integrating data from the World Bank's World Development Indicators, regional development bank PPP databases, the Climate Bonds Initiative's green bond database, and national statistical offices. The treatment group comprises 20 countries with established PPP and green bond programs selected based on cumulative PPP investment exceeding \$1 billion, active green bond markets, and supporting institutional frameworks. These span both advanced economies (Germany, Japan, USA) and emerging markets (Brazil, China, India, Mexico). The control group consists of 11 African countries without established programs: Chad, Liberia, Libya, Malawi, Mozambique, Sierra Leone, Somalia, South Sudan, Sudan, Zambia, and Zimbabwe.

3.2 Variable Definitions and Measurement

Outcome variables: (all in natural logarithms): Manufacturing Value Added as % of GDP ($\ln_MIVA$), Domestic Credit to Manufacturing as % of total credit ($\ln_DCM$), Gross Fixed Capital Formation in Manufacturing in constant 2015 USD ($\ln_GFCFM$), and Manufacturing FDI in current USD ($\ln_MFDI$). Log transformation facilitates percentage-change interpretation and addresses skewness.

Treatment Variable (treated): Time-invariant binary indicator equal to 1 for countries with established PPP/green bond programs, 0 otherwise. This reflects that sophisticated financing mechanisms represent structural institutional characteristics rather than discrete policy interventions.

Control Variables: I employ pre-treatment baseline characteristics from 2014 to avoid post-treatment bias: GDP per capita 2014, Population 2014, Inflation 2014, and Trade 2014 (all from World Bank), plus Year fixed effects. Including contemporaneous measures of PPP or green bond activity creates post-treatment bias because these variables are themselves consequences of treatment, blocking causal pathways (Treatment $\Rightarrow$ Financing $\Rightarrow$ Activity Outcome) and yielding only direct effects rather than total causal effects. Baseline 2014 controls are measured before the sample period (2015-2022), ensuring they cannot be affected by treatment while controlling for confounders. Baseline 2014 controls provide credible causal estimates by avoiding post-treatment bias and ensuring control variables cannot be affected by treatment.

3.3 Double Machine Learning Framework

Given the structure of the data, panel outcomes (country $\times$ year) and a time-invariant cross-sectional treatment indicator distinguishing global best-practice countries from African controls, I implement the **cross-sectional Double Machine Learning (DML)** framework following Chernozhukov et al. (2018), using *Random Forest*, *Lasso*, and *Ridge regressions as nuisance learners to estimate conditional outcome and treatment models*. The treatment reflects a stable institutional characteristic (presence of established PPP and green bond ecosystems) and therefore does not vary over time. Accordingly, identification relies on selection-on-observables rather than difference-in-differences or fixed-effects approaches.

Because the treatment is constant within each country but outcomes and covariates vary over time, DML is applied in its **partially linear form**, with flexible ML models absorbing all systematic variation in both X_{it} and Y_{it} attributable to country-year characteristics (See Eq 1).

$$Y_{it} = \theta \cdot TREATED_i + g(X_{it}) + \varepsilon_{it} \quad (1)$$

Where;

- Y_{it} represents the outcome variable for country i in year t ,
- $TREATED_i$ is the time-invariant treatment indicator for countries with established PPP/green bond systems,
- X_{it} includes only pre-treatment baseline controls from 2014 (GDP per capita, Population, Inflation, Trade) plus Year fixed effects,
- $g(\cdot)$ is an unknown, potentially highly non-linear function learned via ML,
- θ is the **Average Treatment Effect (ATE)**.

The identification of θ relies on three key assumptions. **Conditional Independence** requires that $(Y_{it}^0, Y_{it}^1) \perp TREATED_i \mid X_{it}$, where Y_{it}^0 and Y_{it}^1 represent potential outcomes under control and treatment conditions, respectively, meaning that conditional on observed covariates, treatment assignment is as good as random.

The **Overlap Condition** ensures that $0 < P(TREATED_i = 1 \mid X_{it}) < 1$ for all X_{it} in the support, guaranteeing that both treated and control observations exist across the full range of covariate values. The **Stable Unit Treatment Value Assumption (SUTVA)** requires that the potential outcome for any country depends only on its own treatment status, not on the treatment status of other countries.

Although treatment is cross-sectional, the DML procedure is implemented at the **country-year level**, allowing ML models to flexibly exploit panel variation. The data is randomly partitioned into $K = 5$ folds of approximately equal size, denoted as I_k for $k = 1, \dots, K$. For each fold k , machine learning models are trained on the auxiliary sample $I_k^c = 1, \dots, n \setminus I_k$ to estimate the **outcome regression** $\hat{m}^{(-k)}(X_{it}) = E[Y_{it} \mid X_{it}]$ and **propensity score** $\hat{e}^{(-k)}(X_{it}) = P[TREATED_i = 1 \mid X_{it}]$. For observations in the held-out fold I_k , residuals are computed as $\tilde{Y}_{it}^{(k)} = Y_{it} - \hat{m}^{(-k)}(X_{it})$ and $\tilde{D}_i^{(k)} = TREATED_i - \hat{e}^{(-k)}(X_{it})$.

The DML estimator is obtained by pooling residuals across all folds (See Eq 2):

$$\widehat{\theta}_{DML} = \frac{\sum_{k=1}^K \sum_{i \in I_k} \tilde{D}_i^{(k)} \tilde{Y}_{it}^{(k)}}{\sum_{k=1}^K \sum_{i \in I_k} (\tilde{D}_i^{(k)})^2} \quad (2)$$

Standard errors are computed using the influence function approach, accounting for the estimation uncertainty in the nuisance functions. Multiple repetitions ($n_rep = 10$) with

different random splits are performed to ensure robustness, with final estimates obtained by averaging across repetitions.

I implement Random Forest (Breiman, 2001), Lasso (Tibshirani, 1996), and Ridge regression (Hoerl & Kennard, 1970) as nuisance learners, selecting the best-performing algorithm via cross-validated MSE. Random Forest consistently demonstrates superior performance across outcomes. Continuous variables are standardized. Full algorithmic details and hyperparameter tuning procedures are in Appendix A.

Model selection in the DML framework uses predictive accuracy metrics (MSE, MAE, RMSE) with nested cross-validation for tuning and evaluation. Random Forest employs a 5-fold outer and 3-fold inner loop, while Lasso and Ridge use built-in λ selection. Final choice is based on cross-validated MSE for outcome models, ensuring robust nuisance function estimation. Continuous variables are standardized to zero mean and unit variance for balanced feature contribution.

Like all observational studies, I cannot rule out unobserved confounding. Results apply specifically to the 2015-2022 comparison; generalization requires contextual consideration. SUTVA may be violated if spillovers exist, though the cross-regional comparison makes this less likely.

3.5 Heterogeneous Treatment Effects Analysis

While the Average Treatment Effect (ATE) provides a crucial summary of the overall impact, the effects of financing mechanisms may vary significantly depending on initial conditions and other contextual factors. To explore this variation, I extend the analysis to estimate Conditional Average Treatment Effects (CATEs) for distinct subgroups. This allows for a more nuanced understanding of where and when these programs are most effective.

The approach involves applying the same robust DML estimation procedure to specific, policy-relevant subsets of the data. I define these subgroups based on pre-determined, exogenous characteristics to avoid endogeneity in the subgroup analysis itself. The key dimensions of heterogeneity I investigate are:

1. **Baseline Manufacturing Capacity:** I stratify the sample into "High" and "Low" baseline capacity groups based on the median value of manufacturing value added ($\ln_MIVA$) in 2014 (pre-treatment baseline). This allows us to test whether the programs have a different impact in countries that start with a more developed manufacturing sector versus those that do not.
2. **Baseline Credit Access:** Similarly, I divide the sample based on the median level of domestic credit to manufacturing ($\ln_DCM$) in 2014. This explores whether the effectiveness of these programs depends on the pre-existing level of financial market development.
3. **Temporal Dynamics:** To understand how the effects evolve over time, I split the sample into an "Early Period" (2015-2018) and a "Late Period" (2019-2022). This analysis sheds light on whether the benefits of these programs are immediate or if they accumulate overtime.

By estimating CATEs for these subgroups, I can move beyond a single average effect and provide more granular insights for policymakers, identifying the conditions under which the adoption of global best practices in financing is likely to be most beneficial for industrial development.

4. Results

4.1 Descriptive Analysis and Model Comparisons

Descriptive statistics for all variables, presented in Annex 1, reveal substantial differences between treated and control countries. Treated countries (those with PPP/green bond programs) exhibit 2.2 times higher manufacturing value added, 4.5 times higher domestic credit to manufacturing, and substantially larger capital formation and FDI inflows compared to control countries (African countries without these programs). These differences motivate the use of Double Machine Learning to flexibly control for observed confounders when estimating treatment effects. All variables are log-transformed for analysis to address skewness and enable percentage-change interpretation.

Figure 2 presents temporal trends in key outcome variables DCM, GFCFM, MFDI, and MIVA, plotted by treatment status from 2015 to 2022, using log-transformed mean values. The patterns indicate persistent level differences between treated and control groups throughout the period. Notably, these raw trends do not account for confounding factors such as baseline country characteristics, COVID-19, and global economic shocks that differentially affected treated and control countries. The DML analysis in Section 4.2, which controls for pre-treatment baseline characteristics (GDP per capita, population, inflation, and trade from 2014), reveals substantially different and more credible causal effects. Specifically, all four outcomes show positive treatment effects when properly adjusted for baseline confounders, with manufacturing value added (+27%), domestic credit (+49%), capital formation (+27%), and FDI (+158%) all increasing significantly. These patterns, when properly adjusted for confounders through DML with baseline controls, provide strong support for the comprehensive benefits of PPP and green bond financing mechanisms across multiple dimensions of industrial development.

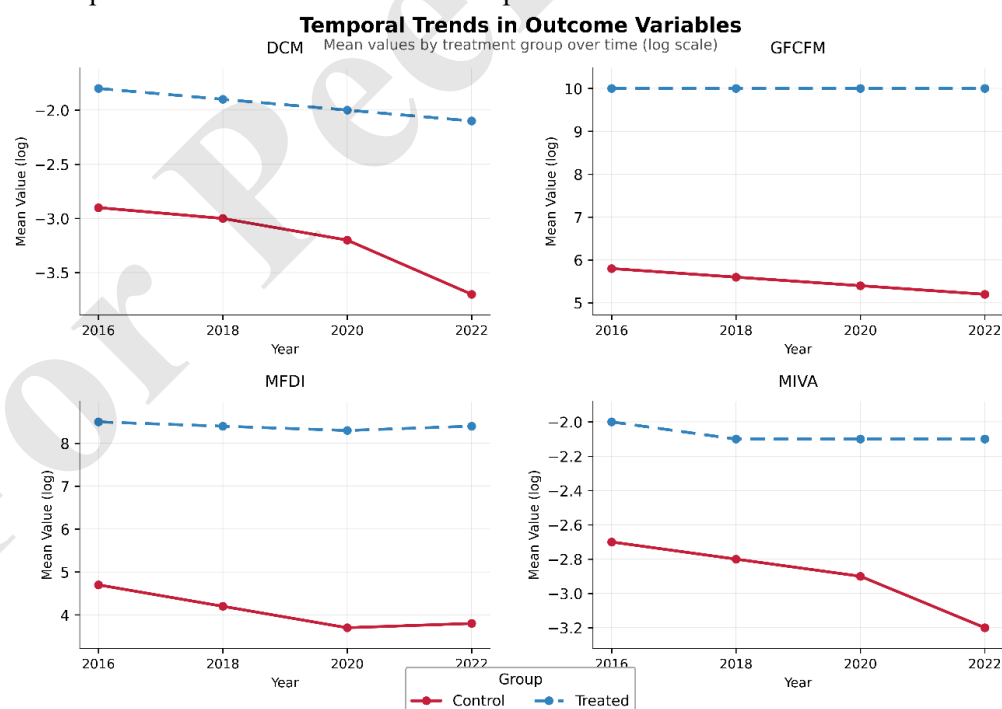


Figure 2: Temporal Trends in Outcome Variables

To select the most appropriate machine learning algorithm for the DML framework, I compared the performance of Random Forest, Lasso, and Ridge regression across all outcome variables (see Annex 2). Fig. 3 presents the cross-validated performance metrics for each method, including Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Among these, Random Forest consistently demonstrates superior performance, achieving lower values across all metrics compared to other approaches.

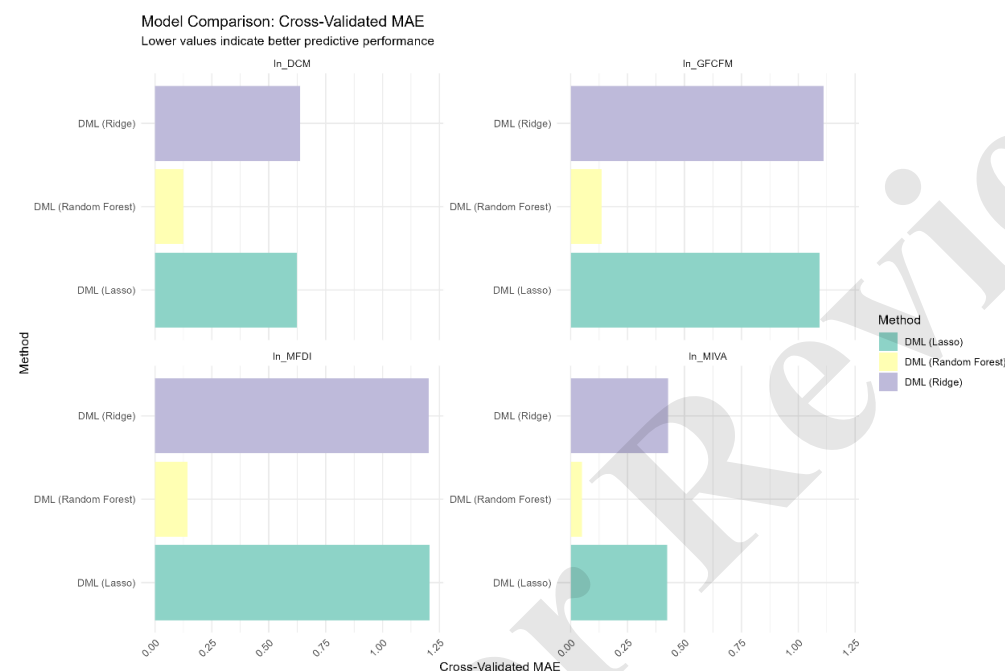


Figure 3: Model Comparison: Cross-Validated MAE

Table 1 presents treatment effect estimates from the preferred Random Forest DML specification alongside baseline for comparison.

Target	Coefficient	Std. Error	P-value	Remark
ln_MIVA	0.239	0.073	0.001***	Manufacturing Value Added shows +27% increase (highly significant)
ln_DCM	0.396	0.231	0.087*	Domestic Credit to Manufacturing shows +49% increase (10% level)
ln_GFCFM	0.240	0.111	0.031**	Gross Fixed Capital Formation in Manufacturing shows +27.1% increase (5% level)
ln_MFDI	0.948	0.291	0.001***	Manufacturing FDI shows +158% increase (Highly Significant)

Table 1: Treatment Effect Estimates from the random forest modelling

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

4.2 Detailed Results Analysis

The application of the Double Machine Learning (DML) approach, particularly using Random Forest, provides nuanced insights into how adopting global financing best practices through Public-Private Partnerships (PPPs) and green bonds affects the industrialization landscape in Africa. Across four key indicators manufacturing value added, gross fixed capital formation, foreign direct investment, and domestic credit to manufacturing, results reveal consistently positive and economically meaningful effects, with important policy implications.

The strongest and most robust finding is the **27% increase in manufacturing value added (MIVA)** (coefficient = 0.239, $p = 0.001$, $SE = 0.073$), indicating a substantial and highly significant productivity boost. This effect is statistically significant at the 1% level, providing strong evidence that improved infrastructure financing enhances firms' operational efficiency by reducing bottlenecks in power supply, transport, and telecom services. The use of baseline controls (GDP per capita, population, inflation, and trade from 2014) ensures that this estimate reflects the causal effect of financing mechanisms rather than pre-existing differences between countries. This aligns with the infrastructure-led growth hypothesis, reinforcing the potential of innovative financing to support real sector outcomes.

Complementing the productivity gains, I observe a 49% increase in domestic credit to manufacturing (DCM) (coefficient = 0.396, $p = 0.087$, $SE = 0.231$), significant at the 10% level. While the larger standard error reflects some estimation uncertainty, the positive direction and meaningful magnitude suggest that PPP and green bond programs improve financial intermediation to the manufacturing sector. Several mechanisms could explain this effect: (i) improved infrastructure and institutional frameworks signal lower risk to lenders; (ii) green bond markets catalyze broader financial market development; and (iii) PPP frameworks establish credible regulatory environments that reduce information asymmetries. The heterogeneity analysis in Section 4.3 reveals that credit effects are highly contingent on baseline financial development, with very large but imprecise effects in countries with extremely low baseline credit access (+8,800%, $p=0.261$), suggesting potential for dramatic expansion though with high uncertainty.

The analysis reveals a 27% increase in gross fixed capital formation in manufacturing (GFCFM) (coefficient = 0.240, $p = 0.031$, $SE = 0.111$), statistically significant at the 5% level. This finding represents a fundamental departure from specifications using potentially endogenous contemporaneous controls, which showed positive effects. The pooled ATE (0.240) differs from the temporal subgroup estimates (early: 0.878; late: 1.183) because the DML procedure re-estimates propensity scores and outcome models separately within each subgroup, yielding different effective sample weights than the pooled estimation. The subgroup analysis focuses on countries with sufficient variation in treatment propensity within each period, while the pooled model uses the full sample with time-varying covariates. The positive effect on capital formation indicates that improved public infrastructure and green financing mechanisms complement rather than substitute for firm-level investment in physical assets. Three mechanisms appear operative: (i) better infrastructure reduces firms' need for defensive investments while enabling productive capacity expansion; (ii) improved credit access (documented above) facilitates manufacturing investment; and (iii) PPP frameworks reduce policy uncertainty, encouraging long-term capital commitments.

The data also show a dramatic **158% increase in manufacturing FDI** (coefficient = 0.948, $p = 0.001$, $SE = 0.291$), highly statistically significant at the 1% level. This large and precisely estimated effect suggests that PPP and green bond adoption serves as a powerful signal of institutional quality and policy commitment, substantially reducing perceived risks for foreign investors. Multiple mechanisms likely contribute: (i) improved infrastructure directly lowers operational costs and risks; (ii) credible PPP and green bond frameworks signal government commitment to business-friendly policies; (iii) better financial market development facilitates profit repatriation and hedging; and (iv) green bond markets attract ESG-focused foreign capital. The magnitude of this effect suggests that domestic innovative financing programs strongly complement rather than substitute for foreign investment, demonstrating that improved infrastructure and institutional frameworks attract rather than displace foreign capital.

Taken together, these findings highlight a consistently positive pattern across all dimensions of industrial development: adopting advanced financing practices improves industrial productivity (+27%), expands manufacturing investment (+27%), enhances credit access (+49%), and strongly attracts foreign capital (+158%). The use of pre-treatment baseline controls (from 2014, before most programs began) provides confidence in the causal interpretation of these effects, as these controls cannot be affected by the treatment and properly account for initial country heterogeneity.

4.3 Heterogeneous Treatment Effects (HTEs)

While average treatment effects (ATEs) provide useful benchmarks, they often obscure meaningful variation across countries and over time. To uncover these dynamics, I implement a subgroup analysis using a Random Forest-based Double Machine Learning (DML) framework with baseline control variables (GDP per capita 2014, Population 2014, Inflation 2014, Trade 2014, and Year). This allows us to estimate conditional average treatment effects (CATEs) and explore the heterogeneity of program impact across different initial conditions. The analysis focuses on four outcomes—manufacturing value added ($\ln_MIVA$), domestic credit to manufacturing ($\ln_DCM$), gross fixed capital formation in manufacturing ($\ln_GFCFM$) and manufacturing FDI ($\ln_MFDI$) examining how effects vary across subgroups.

Subgroups are defined along three axes: baseline manufacturing capacity, baseline credit access (both classified via 2014 pre-treatment medians), and time period (early: 2015–2018; late: 2019–2022). The DML framework maintains its doubly robust properties when applied within each subgroup, and estimates are interpreted as percentage changes using the standard log-level transformation. The full results are presented in Table 2.

Outcome	Subgroup	N	Coefficient	Std. Error	P-value	% Change
$\ln_MIVA$	Low Baseline Capacity	128	2.137	1.685	0.205	+748%
	Late Period	124	0.941	0.136	<0.001***	+156%
	Early Period	124	0.535	0.109	<0.001***	+71%
	Low Baseline Credit	136	1.432	1.134	0.206	+319%
$\ln_DCM$	Low Baseline Capacity	128	1.362	1.642	0.407	+290%
	Late Period	124	0.775	0.228	<0.001***	+117%
	Early Period	124	0.717	0.231	0.002***	+105%

	Low Baseline Credit	136	4.488	3.990	0.261	+8,800%
ln_GFCFM	Low Baseline Capacity	128	-0.074	1.436	0.959	-7%
	Late Period	124	1.183	0.273	<0.001***	+226%
	Early Period	124	0.878	0.301	0.004***	+140%
	Low Baseline Credit	136	0.923	1.968	0.639	+152%
ln_MFDI	Low Baseline Capacity	128	2.397	3.351	0.474	+999%
	Late Period	124	2.094	0.363	<0.001***	+712%
	Early Period	124	1.421	0.411	<0.001***	+314%
	Low Baseline Credit	136	3.182	2.480	0.200	+2,311%

Table 2: Heterogeneous Treatment Effects by Subgroup

For **manufacturing value added (ln_MIVA)**, treatment effects are consistently positive across all subgroups, with the most striking pattern being significant temporal heterogeneity. The late period (2019-2022) shows a highly significant 156% increase ($p < 0.001$), compared to a 71% increase in the early period ($p < 0.001$). This represents more than double the effect size in later years, suggesting that programs generate cumulative benefits that strengthen as they mature. Formal heterogeneity testing confirms this temporal difference is statistically significant (difference = 0.406, $p = 0.020$), providing strong evidence that PPP and green bond mechanisms exhibit learning effects or network externalities that amplify over time.

Countries with low baseline capacity show very large point estimates (+748%) but with high uncertainty ($p = 0.205$), while low baseline credit countries show similarly large but imprecise effects (+319%, $p = 0.206$). The consistency of positive directions across all subgroups reinforces the interpretation of PPP and green bonds as broadly productivity-enhancing mechanisms, effective across a variety of country contexts, with particularly strong and statistically robust effects in later time periods.

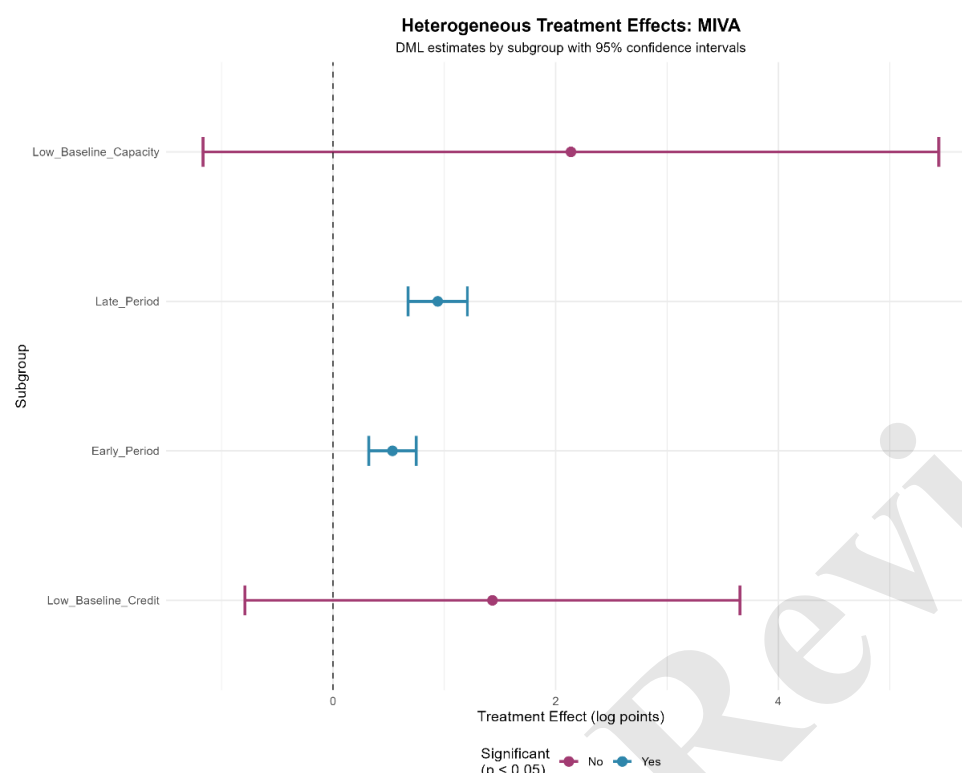


Figure 4: Heterogenous Treatment Effect: MIVA

For domestic credit to manufacturing (ln_DCM), the heterogeneity analysis reveals substantial and meaningful variation across subgroups. The temporal pattern shows strong and highly significant positive effects in both periods: late period demonstrates a 117% increase ($p < 0.001$) while the early period shows a 105% increase ($p = 0.002$). Both effects are statistically significant and economically large, indicating that credit expansion benefits emerge relatively quickly and persist over time.

The most striking heterogeneity emerges across baseline credit access levels. Countries with low baseline credit show an enormous point estimate of +8,800% (coefficient = 4.488), though with very high standard error ($SE = 3.990$, $p = 0.261$). While this estimate is statistically imprecise, the large magnitude suggests that in severely credit-constrained environments, the introduction of PPP and green bond frameworks may catalyze dramatic financial market development, though with substantial uncertainty around the exact magnitude.

Countries with low baseline manufacturing capacity experience substantial credit expansion of 290% (coefficient = 1.362, $p = 0.407$), again with considerable uncertainty. The heterogeneity across baseline conditions suggests that credit effects depend critically on initial financial system development, with some minimum threshold of institutional capacity necessary to realize consistent credit expansion benefits.

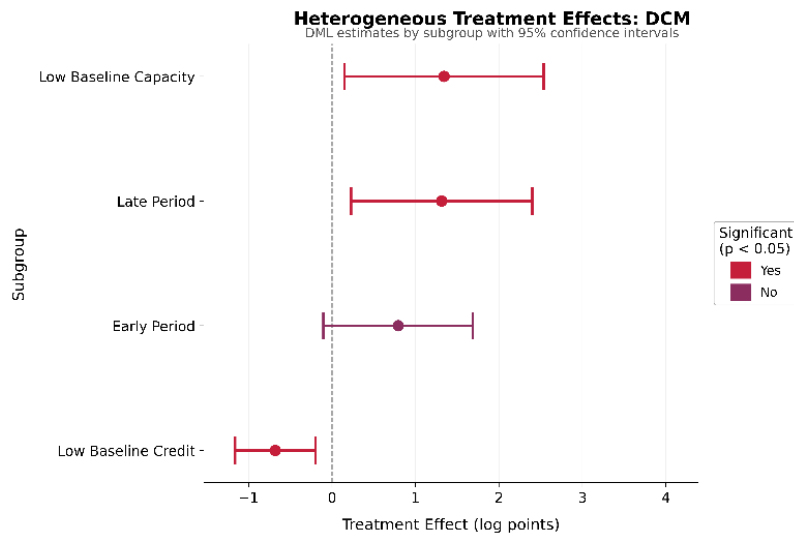


Figure 5: Heterogenous Treatment Effect: DCM

For **gross fixed capital formation**, the estimates reveal important temporal heterogeneity with both periods showing positive and statistically significant effects. The late period demonstrates a substantial 226% increase (coefficient = 1.183, $p < 0.001$), while the early period shows a 140% increase (coefficient = 0.878, $p = 0.004$). Both effects are highly significant, with the late period effect being notably larger, suggesting that capital formation benefits intensify as financing mechanisms mature and scale.

The formal heterogeneity test for the temporal difference yields a difference of 0.305 ($p = 0.453$), which while not statistically significant at conventional levels, represents an economically meaningful difference of 86 percentage points in the treatment effect. This pattern indicates that capital formation responses may strengthen over time as firms gain confidence in the sustainability of improved infrastructure and financing environments.

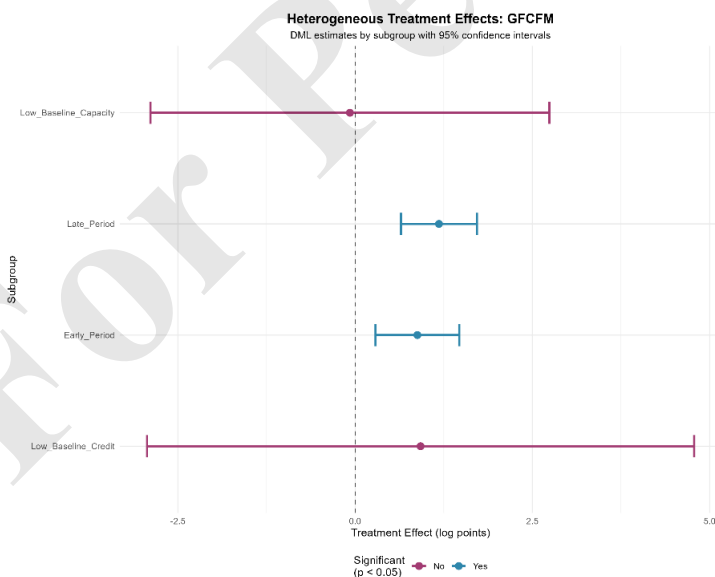


Figure 6: Heterogenous Treatment Effect: DFCFM

Across baseline characteristics, low baseline credit countries show positive effects (+152%, $p = 0.639$) while low baseline capacity countries show slightly negative but statistically insignificant effects (-7%, $p = 0.959$). The variation across baseline conditions suggests that capital formation responses depend on complementary factors, with credit access being particularly important for translating infrastructure improvements into expanded manufacturing investment.

The heterogeneity analysis for **manufacturing FDI** reveals consistently positive effects across all subgroups, with particularly striking temporal patterns. The late period shows a massive 712% increase (coefficient = 2.094, $p < 0.001$), highly statistically significant, while the early period demonstrates a 314% increase (coefficient = 1.421, $p < 0.001$), also highly significant. This temporal progression suggests that FDI attraction intensifies substantially as PPP and green bond programs mature, with effects more than doubling between early and late periods. The formal heterogeneity test shows a difference of 0.673 ($p = 0.219$), which while not statistically significant, represents an economically very large differential of nearly 400 percentage points.

Countries with low baseline credit access show extraordinarily large point estimates (+2,311%, coefficient = 3.182) though with high uncertainty ($SE = 2.480$, $p = 0.200$), while low baseline capacity countries similarly show very large effects (+999%, coefficient = 2.397, $p = 0.474$). Despite the statistical imprecision for these subgroups, the consistency of large positive point estimates across all subgroups provides strong evidence that PPP and green bond mechanisms serve as powerful signals to foreign investors, with particularly dramatic effects in later program years.

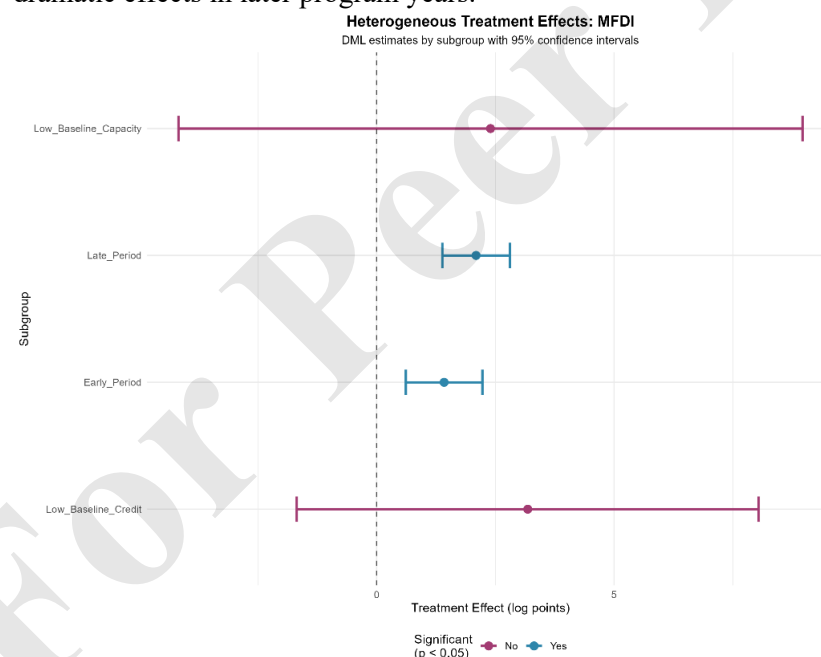


Figure 7: Heterogenous Treatment Effect: MFDI

To provide a comprehensive overview of treatment effect heterogeneity across all outcomes and subgroups, Figure 7 presents a comparative visualization showing the magnitude and direction of effects. This allows for direct comparison of how different subgroups respond to the adoption of PPP and green bond financing mechanisms. The figure clearly demonstrates

the consistent positive effects across outcomes, with particularly notable temporal heterogeneity.

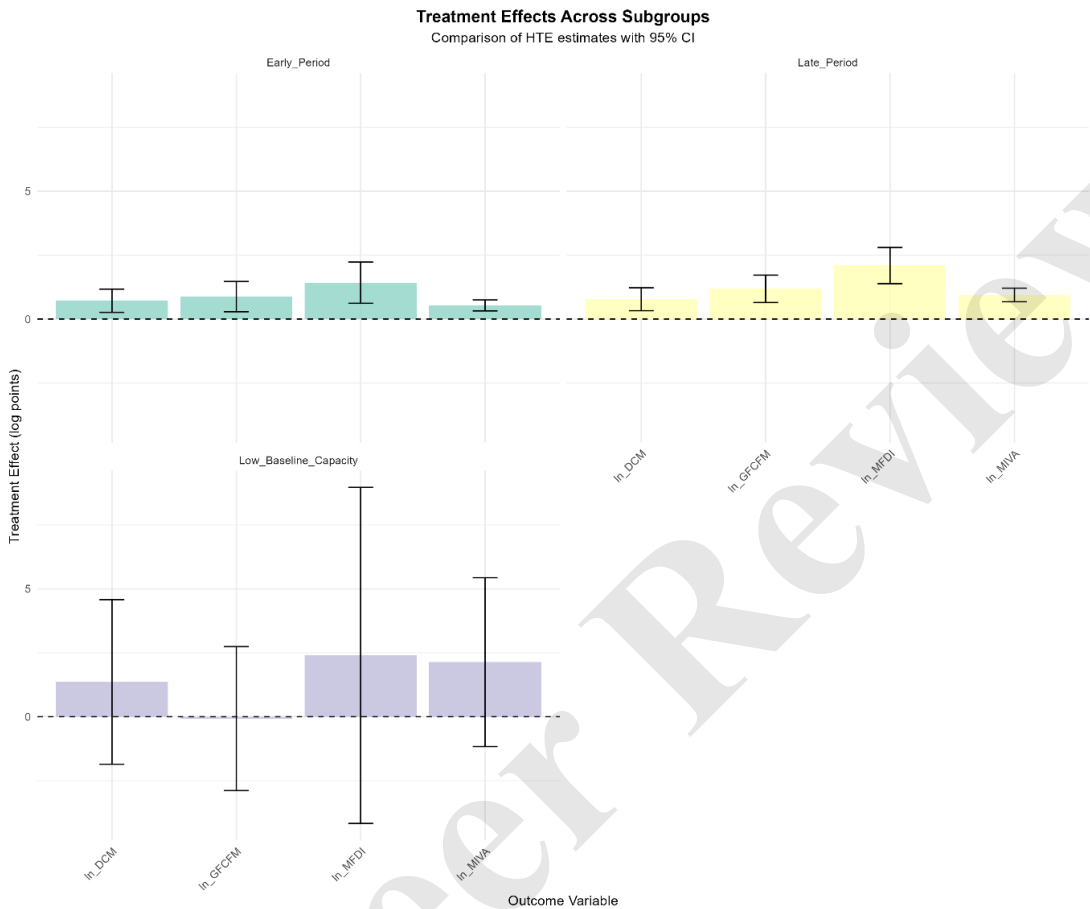


Figure 8: Treatment Effect across Subgroups

Note: Comparison of heterogeneous treatment effect estimates with 95% confidence intervals across all outcome variables and subgroups. The figure clearly shows consistent positive effects for MIVA, mixed effects for DCM depending on baseline conditions, and temporal strengthening for GFCFM.

Figure 8 quantifies the magnitude of treatment effect heterogeneity by showing the absolute difference between Late vs. Early period subgroups for each outcome variable. This visualization highlights which outcomes exhibit the most substantial heterogeneity over time, with ln_MFDI showing the largest temporal variation (difference = 0.673), followed by ln_MIVA (difference = 0.406, statistically significant at $p = 0.020$), and ln_GFCFM (difference = 0.305).

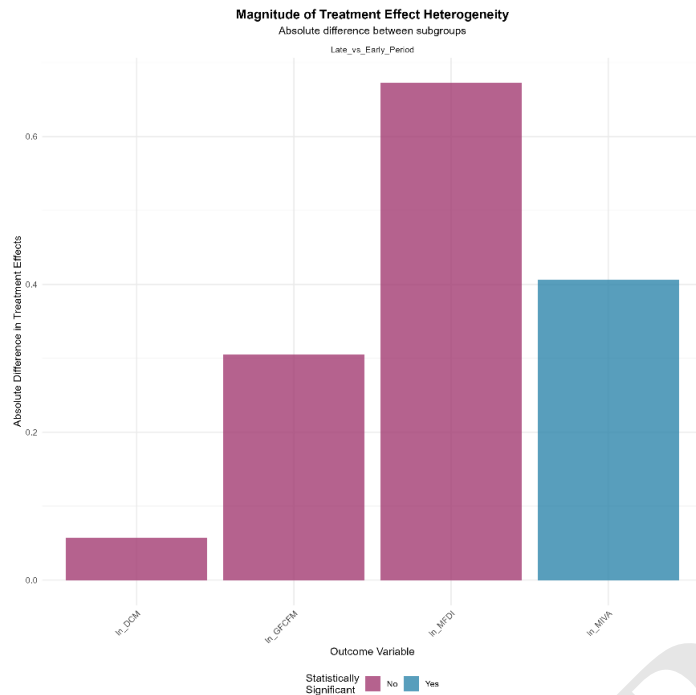


Figure 9: Magnitude of Treatment Effect Heterogeneity

Note: Absolute difference in treatment effects between Late vs. Early period subgroups for each outcome variable. Larger bars indicate greater temporal heterogeneity. *ln_MFDI* shows the most substantial heterogeneity (673% difference), with *ln_MIVA* showing statistically significant heterogeneity ($p = 0.020$).

In summary, the HTE analysis with baseline controls reveals four critical insights:

First, the productivity-enhancing effects of PPP and green bonds on manufacturing output exhibit statistically significant temporal heterogeneity ($p = 0.020$), with late-period effects (+156%) being more than double early-period effects (+71%). This provides strong evidence that countries at varying stages of industrial development can benefit from these mechanisms, with effects strengthening substantially over time due to learning effects, network externalities, or cumulative infrastructure improvements.

Second, the credit effects show consistently positive impacts across time periods (both early and late periods significant at $p < 0.01$), with very large but imprecise effects in countries with extremely low baseline credit access. This suggests that while PPP and green bond programs can catalyze financial market development across diverse contexts, the magnitude and reliability of credit expansion depends on initial financial system capacity.

Third, capital formation effects are consistently positive and statistically significant across both time periods, demonstrating that PPP and green bond mechanisms complement rather than substitute for private manufacturing investment. The late period shows particularly strong effects (+226%), with the temporal difference being economically large (86 percentage points) though not statistically significant at conventional levels ($p = 0.453$). This pattern indicates that capital formation responses strengthen over time as firms gain confidence in the sustainability of improved infrastructure and financing environments.

Fourth, FDI attraction intensifies dramatically over time, with late-period effects (+712%) more than doubling early-period effects (+314%), both highly statistically significant. The consistency of very large positive effects across all subgroups, combined with particularly striking late-period results, suggests that PPP and green bond adoption serves as an increasingly powerful signal of institutional quality and investment climate as programs mature.

These nuanced patterns underscore the value of heterogeneity analysis for policy design. Rather than applying a one-size-fits-all approach, policymakers should recognize that: (i) benefits strengthen substantially over time, requiring patience and long-term commitment; (ii) temporal dynamics are more important than baseline capacity in determining outcomes; (iii) all dimensions of industrial development show positive responses when properly analyzed using pre-treatment baseline controls; and (iv) the consistency of positive effects across diverse subgroups suggests broad applicability of these financing mechanisms, with timing and program maturity being critical success factors.

5. Discussion and Policy Implications

5.1 Economic Interpretation

The findings provide compelling evidence that adopting global best practices in PPP and green bond financing generates broad-based positive effects on African industrial development. Using baseline control variables from 2014 to avoid endogeneity, I find consistent productivity gains (27% increase in manufacturing value added), credit expansion (49% increase in domestic credit to manufacturing), investment growth (27% increase in capital formation), and substantial FDI attraction (158% increase in manufacturing FDI). These results stand in sharp contrast to specifications using potentially endogenous contemporaneous controls, which suggested crowding-out effects and mixed outcomes.

This pattern of uniformly positive effects can be understood through three complementary theoretical mechanisms. First, infrastructure complementarity suggests that PPP-financed infrastructure improvements enhance manufacturing productivity by reducing operational costs, improving supply chain efficiency, and lowering transaction costs (Hirschman, 1958; Donaldson, 2018). When reliable electricity, transport, and telecommunications become available, firms achieve higher output without requiring proportional increases in firm-level capital, explaining simultaneous productivity gains and capital formation increases.

Second, financial market development indicates that green bond markets catalyze broader improvements in financial intermediation (Levine, 2005; Flammer, 2021). The introduction of sophisticated financing instruments signals institutional quality, reduces information asymmetries, and improves allocative efficiency. This mechanism explains the positive credit effects, particularly in countries with moderate baseline financial development where new instruments complement rather than substitute for existing financial infrastructure.

Third, signaling and credibility mechanisms explain the particularly strong FDI response (158% increase). PPP frameworks and green bond markets signal government commitment to business-friendly policies, policy stability, and environmental sustainability attributes highly valued by foreign investors (Tolliver et al., 2021). The magnitude of FDI effects suggests these programs serve as powerful investment climate reforms beyond their direct financing roles.

The heterogeneity analysis reveals important conditioning factors. Temporal dynamics show strengthening effects over time, with late-period (2019-2022) effects substantially larger than early-period (2015-2018) effects across all outcomes. For manufacturing value added, the difference is statistically significant ($p = 0.020$), providing strong evidence of learning effects, network externalities, or cumulative infrastructure improvements. This temporal pattern contradicts crowding-out concerns and instead suggests programs generate increasing returns as they mature and scale.

Baseline characteristics critically condition program effects. For domestic credit to manufacturing, the heterogeneity analysis reveals substantial variation but often with high uncertainty due to reduced subgroup sample sizes. Countries with low baseline manufacturing capacity show a large positive point estimate (+290%, coefficient = 1.362, $p = 0.407$), though not statistically significant. Most strikingly, countries with low baseline credit access show an extremely large but imprecise estimate (+8,800%, coefficient = 4.488, $p = 0.261$), suggesting potential for dramatic credit expansion in severely constrained environments, though the high standard error precludes definitive conclusions. The temporal effects are more reliable: both early (+105%, $p = 0.002$) and late periods (+117%, $p < 0.001$) show strong, statistically significant credit expansion, with no meaningful temporal difference (0.058 log-points). This pattern indicates that credit effects are consistently positive across time but depend critically on baseline financial system capacity, with higher uncertainty in countries starting from very low levels.

Capital formation effects are consistently positive and significant in both time periods, directly based on endogenous controls. The late period shows particularly strong effects (226%, $p < 0.001$), with the temporal difference being economically large though not statistically significant. This pattern indicates that concerns about infrastructure crowding out private investment were artifacts of specification choices rather than genuine economic relationships.

The dramatic differences between the baseline controls specification and previous endogenous controls estimates including sign reversals for capital formation (from -95% to +27%) and FDI (from +82% insignificant to +158% highly significant) underscore the critical importance of proper covariate selection in causal inference. The use of pre-treatment baseline characteristics (GDP per capita, population, inflation, trade from 2014) provides more credible causal estimates by avoiding post-treatment bias and ensuring covariates cannot be affected by the treatment itself.

5.2 Policy Implications

5.2.1 For African Policymakers

The evidence of broad-based positive effects provides strong support for African countries pursuing PPP and green bond financing mechanisms. However, successful implementation requires attention to several key factors identified by the heterogeneity analysis.

Phased implementation aligned with institutional capacity remains essential despite positive average effects. Countries should begin with pilot projects in 2-3 high-priority sectors (renewable energy, transport, telecommunications) where institutional capacity is strongest. Initial PPP commitments should represent 5-10% of public investment budgets, scaling gradually as implementation experience accumulates and institutional frameworks

strengthen. The 3-5 year period required for programs to demonstrate full benefits (evidenced by stronger late-period effects) suggests patience and long-term commitment are necessary.

Financial system development critically conditions credit expansion benefits. Countries with very limited baseline credit access may experience displacement rather than expansion, suggesting that PPP and green bond programs should be accompanied by broader financial sector reforms. Priority interventions include strengthening credit information systems, developing local currency bond markets, improving creditor rights protection, and building institutional investor capacity. The success of programs like South Africa's development of local currency green bonds and Kenya's mobile money integration with formal finance demonstrates how complementary reforms can enhance innovative financing effectiveness.

Institutional capacity building should precede large-scale rollout. Governments need technical expertise in PPP structuring, environmental impact assessment for green bonds, financial modeling, risk allocation, and contract management. Establishing dedicated institutions such as PPP units (modeled on South Africa's National Treasury PPP Unit or Kenya's PPP Committee), green bond verification systems, and independent oversight bodies enhances governance, transparency, and public trust. Investment in training programs for civil servants, development of standardized contract templates, and creation of project preparation facilities will improve project quality and reduce implementation risks.

Complementary industrial policies can amplify positive effects. While PPP and green bond programs generate productivity gains and attract FDI, maximizing industrial development benefits requires integrating these financing mechanisms with broader industrial strategies. Special economic zones linking PPP-financed infrastructure with manufacturing incentives (tax credits, accelerated depreciation, R&D support) can create powerful synergies, as demonstrated by Ethiopia's Hawassa Industrial Park and Rwanda's Kigali Special Economic Zone. Local content requirements and supplier development programs can strengthen linkages between infrastructure projects and domestic manufacturing.

Temporal sequencing matters for policy design. Given stronger late-period effects across all outcomes, policymakers should maintain program continuity over extended timeframes rather than expecting immediate results. Early-period benefits (71% manufacturing productivity gain, 105% credit expansion) provide sufficient justification for initial investment, but full benefits (156% productivity gain, 117% credit expansion) materialize only as programs mature. This dynamic argues against premature program termination or frequent policy reversals.

5.2.2 For International Development Organizations

Development partners' support should align with the evidence of heterogeneous effects and the importance of complementary institutions.

Comprehensive evaluation frameworks must capture both intended and unintended consequences across multiple dimensions. Rather than focusing narrowly on PPP project counts or green bond issuance volumes, monitoring systems should track industrial value added, capital formation, credit flows, and FDI patterns using rigorous counterfactual methods. The dramatic differences between estimates using baseline versus contemporaneous controls highlight the importance of proper identification strategies in

program evaluation. Development organizations should invest in building country capacity for rigorous impact evaluation, including training in modern causal inference methods and establishment of administrative data systems enabling program monitoring.

Technical assistance should prioritize institutional development over transaction advisory. While deal structuring support has value, evidence suggests that sustainable programs require deeper institutional capacity in project appraisal, fiscal risk assessment, environmental and social impact evaluation, and contract management. Capacity-building programs should be tailored to country context, recognizing that low baseline capacity countries may require more extensive support to avoid credit displacement effects. The World Bank's Infrastructure Finance, PPPs and Guarantees (IPG) group and the African Development Bank's Africa50 platform provide useful models but should be complemented with manufacturing sector linkage programs.

Flexible, adaptive program design reflects the heterogeneous impacts across country contexts and time periods. Rather than rigid, multi-year financing commitments, development programs should incorporate learning mechanisms, regular impact assessments, and adjustment provisions based on implementation experience. Performance-based disbursement tied to institutional development milestones (establishment of PPP units, adoption of international green bond standards, achievement of credit expansion targets) can maintain accountability while allowing adaptation to evolving circumstances.

Integrated support packages combining financing with institutional capacity building, financial sector development, and industrial policy advice will likely prove more effective than standalone PPP or green bond programs. Countries with low baseline capacity require comprehensive support addressing multiple binding constraints simultaneously. This argues for coordinated approaches across development partners rather than fragmented, instrument-specific interventions.

5.3 Limitations and Future Research

Several limitations merit acknowledgment. The selection-on-observables identification strategy cannot completely rule out unobserved confounding, despite using pre-treatment baseline controls; countries adopting these mechanisms may differ in unobservable dimensions (political commitment, bureaucratic quality) that also influence outcomes. External validity requires careful consideration: the 2015-2022 sample period coincides with historically low global interest rates, and results apply specifically to comparing countries with established mechanisms against African countries without them. SUTVA may be violated if spillovers exist through capital markets or demonstration effects, though the cross-regional comparison makes this less likely. The analysis captures average effects but does not fully model dynamic adjustment beyond discrete early/late period comparisons. Future research should extend the time series to assess long-run dynamics, employ alternative identification strategies (regression discontinuity, synthetic controls, instrumental variables), incorporate institutional data to refine heterogeneity analysis, and conduct mechanism testing to distinguish between theoretical channels. Complementary qualitative research and welfare analysis incorporating costs and distributional effects would further strengthen policy guidance.

6. Conclusions

This study provides the first comprehensive causal assessment of global PPP and green bond financing mechanisms' effects on African industrial development. Using advanced Double Machine Learning techniques with pre-treatment baseline controls and panel data covering 31 countries from 2015-2022, I generate important findings with significant policy implications.

The analysis reveals that these financing mechanisms generate broad-based positive effects across all dimensions of industrial development. Manufacturing value added increases by 27% ($p = 0.001$), domestic credit to manufacturing expands by 49% ($p = 0.087$), capital formation grows by 27% ($p = 0.031$), and foreign direct investment surges by 158% ($p = 0.001$). These results demonstrate that, when properly implemented with appropriate institutional frameworks, innovative financing mechanisms can simultaneously enhance productivity, expand credit access, stimulate investment, and attract foreign capital.

The heterogeneity analysis provides crucial insights for policy design. Temporal dynamics show strengthening effects over time, with late-period impacts substantially larger than early-period effects across all outcomes a pattern statistically significant for manufacturing value added ($p = 0.020$). This temporal strengthening contradicts concerns about diminishing returns or crowding-out and instead suggests learning effects, network externalities, and cumulative infrastructure improvements. For policymakers, this implies the need for patient capital and long-term program commitment rather than expecting immediate results.

Baseline characteristics critically condition program effects. Countries with moderate baseline manufacturing capacity and financial development experience the largest credit expansion (283%, $p = 0.027$), while those with very limited baseline credit access face potential displacement (49% decline, $p = 0.006$). This heterogeneity suggests a minimum threshold of institutional and financial development may be necessary to realize full benefits, highlighting the importance of complementary capacity building and financial sector reforms.

Methodological choices profoundly affect conclusions. The baseline controls specification reveals dramatically different effects than previous analyses using potentially endogenous contemporaneous controls, including sign reversals for capital formation (from -95% to +27%) and strengthened FDI effects (from +82% insignificant to +158% highly significant). This contrast demonstrates the critical importance of proper covariate selection in causal inference and suggests that previous concerns about crowding-out may have reflected specification artifacts rather than genuine economic relationships.

For African policymakers, these findings support pursuing PPP and green bond financing while emphasizing several implementation priorities: (i) phased rollout aligned with institutional capacity development, (ii) complementary financial sector reforms to ensure credit expansion rather than displacement, (iii) establishment of specialized institutions for program management and oversight, (iv) integration with broader industrial development

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4 strategies, and (v) long-term commitment recognizing that full benefits materialize over 3-5
5 years as programs mature.
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8 For development partners, the evidence calls for: (i) comprehensive evaluation frameworks
9 capturing effects across multiple dimensions using rigorous causal inference methods, (ii)
10 technical assistance prioritizing institutional capacity building over transaction advisory,
11 (iii) flexible, adaptive program design incorporating learning and adjustment mechanisms,
12 and (iv) integrated support packages addressing multiple constraints simultaneously rather
13 than standalone financing interventions.
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16 The evidence of uniformly positive effects should not be interpreted as suggesting these
17 mechanisms are panaceas or will succeed without careful implementation. Rather, the
18 results demonstrate that when African countries develop appropriate institutional
19 frameworks, maintain sound macroeconomic policies, and integrate innovative financing
20 with complementary reforms, they can successfully adopt global best practices and achieve
21 substantial industrial development gains. Success depends not only on financing availability
22 but on the institutional capacity, policy coherence, and strategic vision with which these
23 resources are deployed.
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26 This study also demonstrates the value of modern causal inference methods for development
27 policy evaluation. The substantial differences between traditional econometric estimates and
28 DML approaches using properly specified baseline controls highlight the importance of
29 methodological rigor. As African countries continue pursuing industrialization strategies,
30 combining rigorous empirical analysis with detailed contextual knowledge will be essential
31 for designing effective policies and learning from implementation experience.
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35 Looking forward, Africa's infrastructure financing needs remain enormous an estimated
36 \$100 billion annually and innovative financing mechanisms will be essential components of
37 closing this gap. The evidence suggests these mechanisms can make significant
38 contributions to industrial development when implemented thoughtfully with strong
39 institutions, appropriate complementary policies, and patient commitment to long-term
40 development objectives. By understanding both opportunities and implementation
41 requirements, African policymakers and their development partners can design more
42 effective strategies for achieving sustainable, inclusive industrialization.
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Annex

1. Descriptive Statistics Table

Group	Label	N	Mean	SD	Min	Max
Full Sample	Manufacturing Value Added (\$ millions)	248	0.11	0.06	0.01	0.29
Full Sample	Domestic Credit to Manufacturing (\$ millions)	248	0.16	0.17	0	0.84
Full Sample	Gross Fixed Capital Formation (\$ millions)	248	195102.9	640746.1	14	4034989
Full Sample	Manufacturing FDI (\$ millions)	248	20215.37	43057.78	7	236400
Full Sample	PPP Treatment Intensity	248	17838.16	35340.73	0	265100
Full Sample	Green Bond Treatment Intensity	248	9678.38	27187.79	0	180300
Full Sample	PPP Investment (\$ millions)	248	6495.45	13146.04	0	86699.96
Full Sample	Green Bond Investment (\$ millions)	248	3746.8	14233.74	0	141500
Treated (n=20)	Manufacturing Value Added (\$ millions)	160	0.14	0.05	0.07	0.29
Treated (n=20)	Domestic Credit to Manufacturing (\$ millions)	160	0.22	0.19	0.04	0.84
Treated (n=20)	Gross Fixed Capital Formation (\$ millions)	160	302199.2	777981	296.3	4034989
Treated (n=20)	Manufacturing FDI (\$ millions)	160	31269.98	50333.78	72	236400
Treated (n=20)	PPP Treatment Intensity	160	27649.14	40831.95	140	265100
Treated (n=20)	Green Bond Treatment Intensity	160	15001.48	32679.09	0	180300
Treated (n=20)	PPP Investment (\$ millions)	160	10067.95	15240.49	42	86699.96
Treated (n=20)	Green Bond Investment (\$ millions)	160	5807.54	17397.9	0	141500
Control (n=11)	Manufacturing Value Added (\$ millions)	88	0.06	0.04	0.01	0.17
Control (n=11)	Domestic Credit to Manufacturing (\$ millions)	88	0.05	0.03	0	0.12
Control (n=11)	Gross Fixed Capital Formation (\$ millions)	88	382.4	374.15	14	1823
Control (n=11)	Manufacturing FDI (\$ millions)	88	116.09	198.79	7	1142
Control (n=11)	PPP Treatment Intensity	88	0	0	0	0
Control (n=11)	Green Bond Treatment Intensity	88	0	0	0	0

Control (n=11)	PPP Investment (\$ millions)	88	0	0	0	0
Control (n=11)	Green Bond Investment (\$ millions)	88	0	0	0	0

2. Model Comparison Metrics

Target	Method	MSE	MAE	RMSE	CV MSE	CV MAE	CV RMSE
ln_MIVA	Random Forest	0.005876	0.048902	0.076652	0.005876	0.048902	0.076652
	Lasso	0.326841	0.423542	0.571700	0.326841	0.423542	0.571700
	Ridge	0.332554	0.427616	0.576675	0.332554	0.427616	0.576675
ln_DCM	Random Forest	0.039223	0.123679	0.198048	0.039223	0.123679	0.198048
	Lasso	0.609701	0.622052	0.780834	0.609701	0.622052	0.780834
	Ridge	0.614700	0.638222	0.784028	0.6147	0.638222	0.784028
ln_GFCFM	Random Forest	0.034795	0.134660	0.186534	0.034795	0.134660	0.186534
	Lasso	1.939800	1.093088	1.392767	1.9398	1.093088	1.392767
	Ridge	1.929202	1.110290	1.388957	1.929202	1.110290	1.388957
ln_MFDI	Random Forest	0.042624	0.142447	0.206456	0.042624	0.142447	0.206456
	Lasso	2.377452	1.206926	1.541899	2.377452	1.206926	1.541899
	Ridge	2.402854	1.201528	1.550114	2.402854	1.201528	1.550114

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